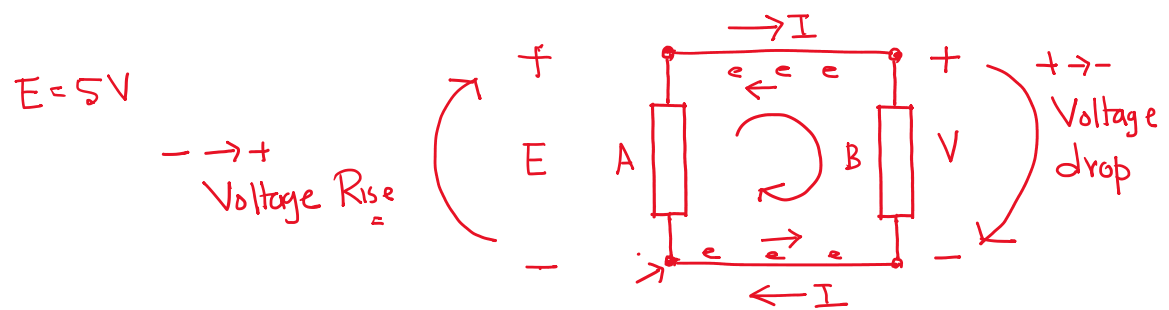




Power delivered by Active element (A)
= Power Absorbed by the passive element (B)

$$P_{del} = -EI \quad P_{abs} = +VI$$

$$P_{del} = P_{abs} \Rightarrow -EI = +VI \Rightarrow -E = V$$

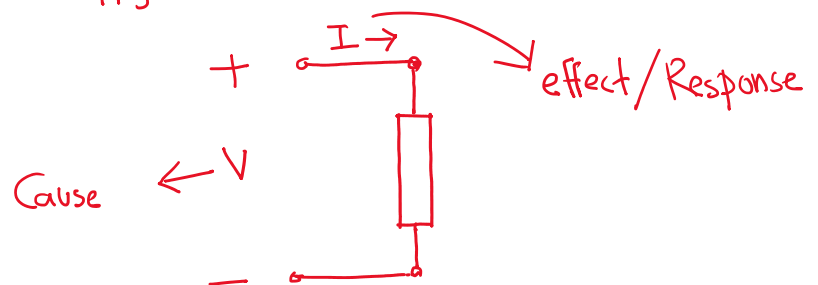
$$|-E| = |V|, \quad V = E$$

Voltage across the passive element: V , is equal to the voltage across active element: E

$$\begin{aligned} E &= V \\ \Rightarrow +E - V &= 0 \\ +E \uparrow - V \downarrow &= 0 \end{aligned}$$

Algebraic sum of all the Voltage around a ckt is zero. (KVL)

Voltage: low $\rightarrow$ to - High: Rise
High $\rightarrow$ to $\rightarrow$ low: Drop

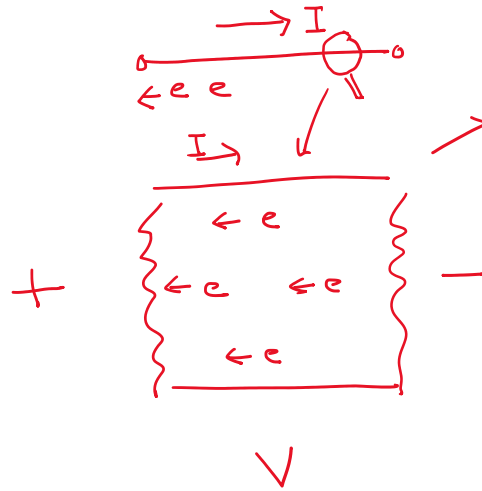


Cause $\rightarrow$ Response

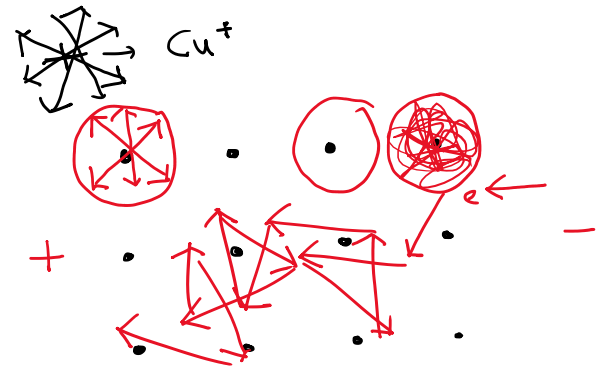
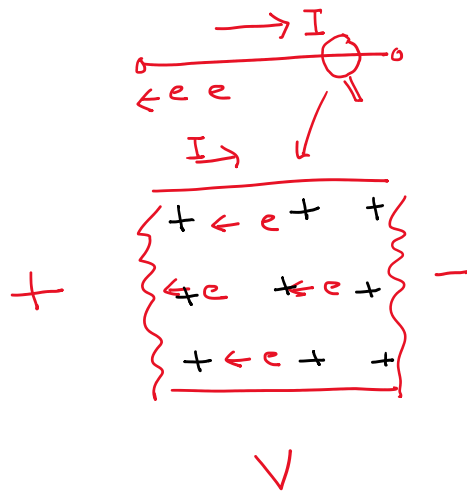
Q: What is the level of the current I For a given potential diff. V .

$V = 10V$, what is the value of I ?
is that $1A$, $2A$, $10A$?

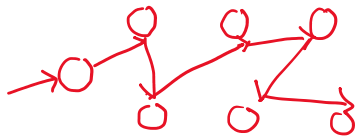
What determines the flow of current ?? Resistance



We assume that the movement of electrons through conductor is straight forward.



electrons scattered by the highly charged Cu-ion.
movement of the electrons experiences opposition



Response/Effect = $\frac{\text{Cause}}{\text{Opposition}} \rightarrow$ measurable entity. R ohm (Ω)

Current = $\frac{\text{Voltage difference}}{\text{Resistance}}$

$$I = \frac{V}{R} \Rightarrow R = \frac{V}{I} \quad (\text{Voltage/Amp})$$

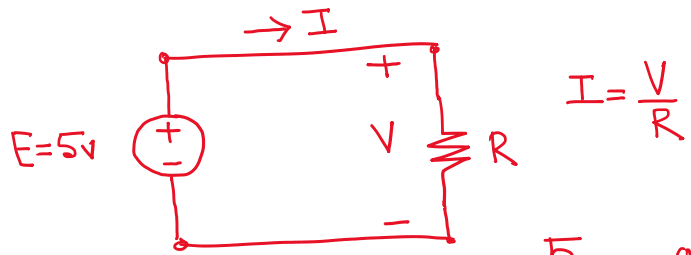
$$I = \frac{V}{R} \Rightarrow V = IR$$

$$R = 1 \Omega = \frac{1V}{1A}$$

Ohm's law



Ohm's law 

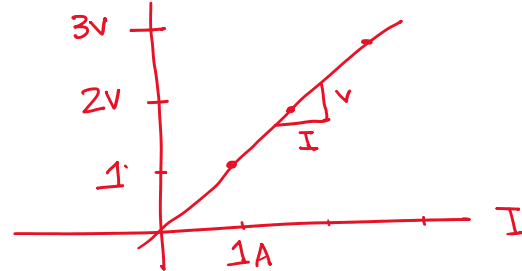


$$R = 10\Omega, \quad I = \frac{5}{10} = 0.5A$$

$$5\Omega, \quad I = \frac{5}{5} = 1A$$

$$V = IR \Rightarrow V \propto I$$

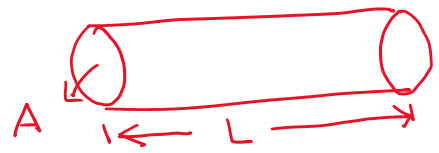
$$V = IR$$



Au, Cu, Al.

$$R \propto L$$

$$\propto \frac{1}{A}$$



$$R = \rho \frac{L}{A} \quad \rho: \text{Resistivity}$$

$$\text{Power} = VI = IR \times I = I^2 R$$

$$P = \frac{1}{\sigma}$$

ρ : Resistivity $\rightarrow$ Ability to resist or oppose the flow of charge

$$\sigma = \frac{1}{\rho}$$

σ : Conductivity $\rightarrow$ Ability to conduct or flow of charge



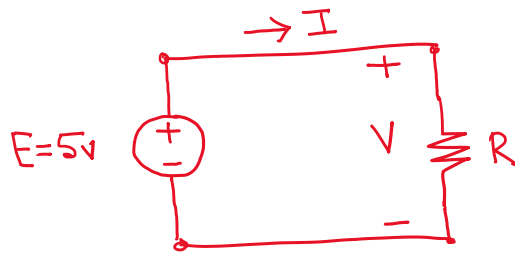
$$I = \frac{V}{R} = V \times \frac{1}{R} = VG$$

$$\boxed{I = VG}$$

$R \leftrightarrow \text{Conductance (G)}$

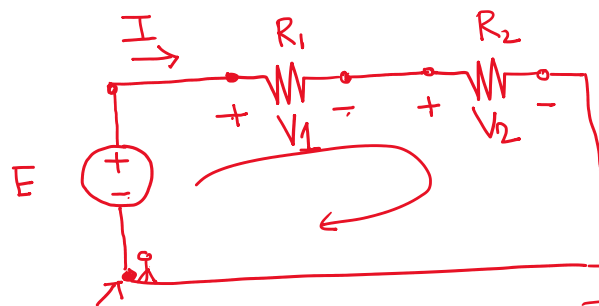
$$G = \frac{1}{R} \text{ (ohm)}$$

(mho)
↓
Siemens



$$I = \frac{V}{R} \text{ (ohm's law)}$$

$$= \frac{E}{R} = \frac{5}{5} = 1 \text{ A}$$



$$R_1 = 10 \Omega$$

$$R_2 = 15 \Omega$$

$$E = 50 \text{ V}$$

$$I = \frac{E}{R_1} / \frac{E}{R}$$

$$\text{KVL: } +E - V = 0 \quad [\text{Kirchoff voltage law}]$$

KCL
current law

Algebraic sum of all the Voltage around a ckt is zero. (KVL)

$$L \rightarrow H: +V_e$$

$$H \rightarrow L: -V_e$$

$$+E - V_1 - V_2 = 0$$

$$E = V_1 + V_2$$

